Computer Networking: A Top-Down Approach
9th Edition (2025)

Chapter 1 – Computer Networks and the Internet
Homework Problems and Questions

Chapter 1 Problems

P5. Review the car-caravan analogy in Section 1.4. Assume a propagation speed of 100 km/hour.
a. Suppose the caravan travels 175 km, beginning in front of one tollbooth, passing through a second tollbooth, and finishing just after a third tollbooth. What is the end-to-end delay?
b. Repeat (a), now assuming that there are eight cars in the caravan instead of ten.

P6. This elementary problem begins to explore propagation delay and transmission delay, two central concepts in data networking. Consider two hosts, A and B, connected by a single link of rate R bps. Suppose that the two hosts are separated by m meters, and suppose the propagation speed along the link is s meters/sec. Host A is to send a packet of size L bits to Host B.
a. Express the propagation delay, dprop, in terms of m and s.
b. Determine the transmission time of the packet, dtrans, in terms of L and R.
c. Ignoring processing and queuing delays, obtain an expression for the end-to-end delay.
d. Suppose Host A begins to transmit the packet at time t = 0. At time t = dtrans, where is the last bit of the packet?
e. Suppose dprop is greater than dtrans. At time t = dtrans, where is the first bit of the packet?
f. Suppose dprop is less than dtrans. At time t = dtrans, where is the first bit of the packet?
g. Suppose $s = 2.5 \times 10^8$ m/s, L = 1500 bytes, and R = 10 Mbps. Find the distance m so that dprop equals dtrans.

P8. Suppose users share a 10 Mbps link. Also suppose each user requires 200 kbps when transmitting, but each user transmits only 10 percent of the time. (See the discussion of packet switching versus circuit switching in Section 1.3.)

a. When circuit switching is used, how many users can be supported?

P10. Consider a packet of length L that begins at end system A and travels over three links to a destination end system. These three links are connected by two packet switches. Let di, si, and Ri denote the len gth, propagation speed, and the transmission rate of link i, for i = 1, 2, 3. The packet switch delays each packet by dproc. Assuming no queuing delays, in terms of di, si, Ri (i = 1, 2, 3), and L, what is the total end-to-end delay for the packet?

P12. A packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, one other packet is halfway done being transmitted on this outbound link and four other packets are waiting to be transmitted. Packets are transmitted in order of arrival. Suppose all packets are 1,500 bytes and the link rate is 2.5 Mbps. What is the queuing delay for the packet?

P13.
a. Suppose N packets arrive simultaneously to a link at which no packets are currently being transmitted or queued. Each packet is of length L and the link has transmission rate R. What is the average queuing delay for the N packets?

b. Now suppose that N such packets arrive to the link every NL/R seconds. What is the average queuing delay of a packet?

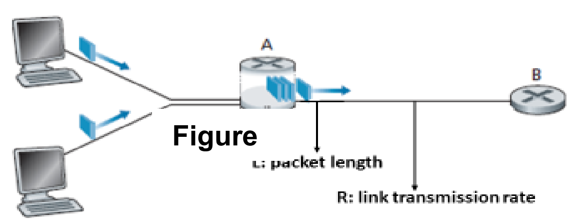
P14. Consider the queuing delay in a router buffer. Let I denote traffic intensity; that is, I = La/R. Suppose that the queuing delay takes the form IL/[R(1 − I)] for I < 1.
a. Provide a formula for the total delay, that is, the queuing delay plus the transmission delay.
b. Plot the total delay as a function of La/R.

P20. Consider the throughput example corresponding to Figure 1.20(b). Now suppose that there are M client-server pairs rather than 10. Denote Rs, Rc, and R for the rates of the server links, client links, and network link. Assume all other links have abundant capacity and that there is no other traffic in the network besides the traffic generated by the M client-server pairs. Derive a general expression for throughput in terms of Rs, Rc, R, and M.

P24. Suppose you would like to urgently deliver 50 terabytes data from Boston to Los Angeles. You have available a 100 Mbps dedicated link for data transfer. Would you prefer to transmit the data via this link or instead use FedEx over-night delivery? Explain.

1. Consider figure 1, in which a single router is transmitting packets, each of length L bits, over a single link with transmission rate R Mbps to another router at the other end of the link. Suppose that the packet length is L= 4000 bits and that the link transmission rate along the link to the router on the right is R = 100 Mbps.

a) What is the transmission delay?
b) Assume the average rate of packets/second (a) is 30. What will be traffic intensity?



2. Consider Figure 1, in which a client sends an http request to a server. The path from client to server has three links, of rates R1 = 100 Mbps, R2 = 10 Mbps, and R3 = 1 Mbps, each with link length of 3km. Suppose that the packet length is L= 18000 bits and speed of light propagation delay on each link is $3x10^8$ m/sec.

a) What is the transmission delay of R2?
b) What is the propagation delay of R1?
c) Name the sources of packet delay?
d) What layer in the IP stack moves datagrams from the source host to the destination?

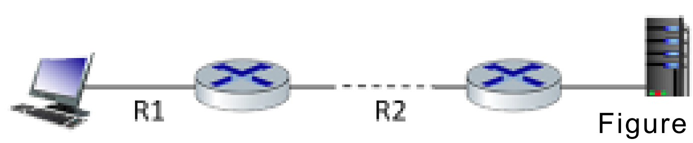
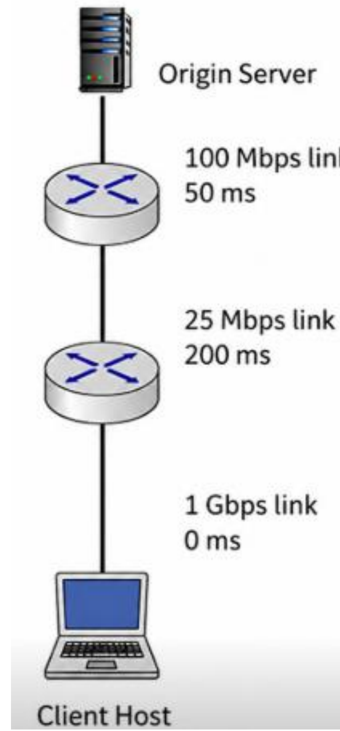


Figure 1

3. Consider the network shown in **Figure**.



An origin server is connected to Router 1 through a 100 Mbps link with a 50 ms propagation delay. Router 1 is connected to Router 2 through a 25 Mbps link with a 200 ms propagation delay. Router 2 is connected to the client host through a 1 Gbps link with a 0 ms propagation delay.

All packets are 20,000 bits long. Assume there is no queuing delay, no processing delay, and no cache.

a) What is the end-to-end delay from the time the packet is transmitted by the server until it is received by the client?

b) What is the maximum rate at which the server can deliver data to a single client, assuming no other clients are making requests?

4. Consider the network shown in the figure. A **packet of size 2,500 bytes** is sent from the **Origin Server** to the **Client Host** through Router 1 and Router 2.

a) Identify the **bottleneck link** in the network. Justify your answer.

b) If traffic on the bottleneck link increases, what happens to the **queueing delay**?

c) If the bottleneck link is upgraded, how can this affect the **end-to-end throughput**?

d) Calculate the **total end-to-end delay** experienced by the packet. Include the **transmission delay and propagation delay** of all links. Assume **no processing delay and no queueing delay**.